



CY 104L – Intro to Cyber

Lab # 08

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Student Name: Athar Imran

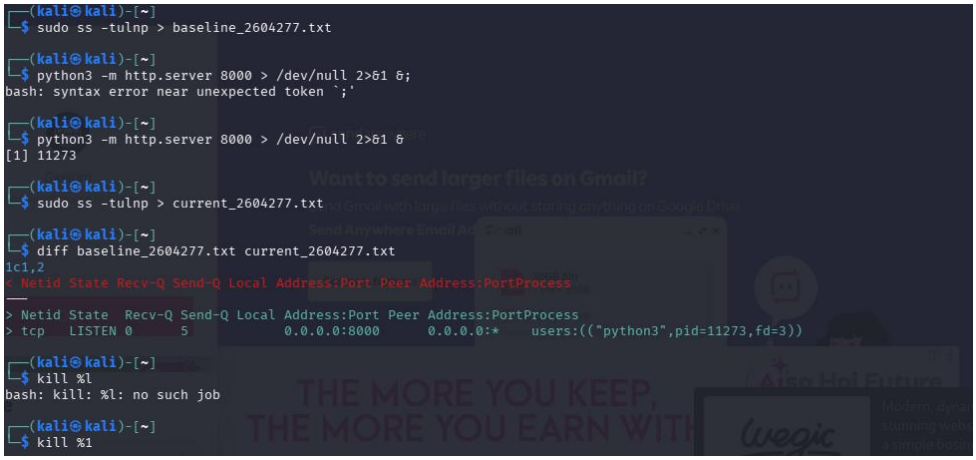
Student ID: 2604277

Assignment Date: 28 Oct, 2026

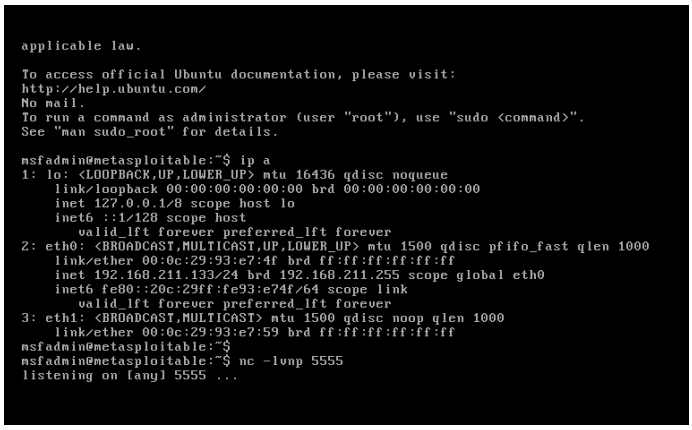
Assignment done: 28 Oct, 2026

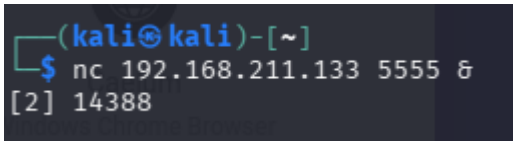
Due Date: 28 Oct, 2026

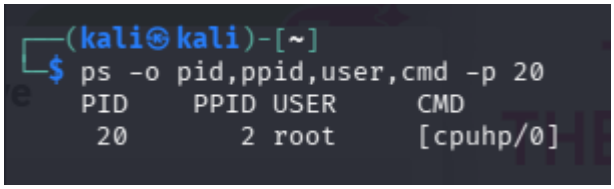
Task 1 – Endpoint Port Baseline

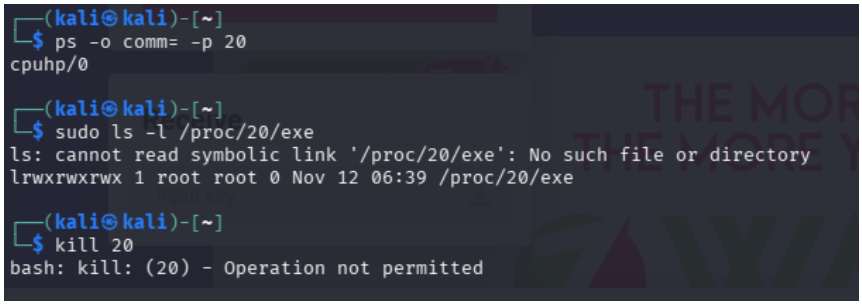


Task 2 – Hunt an Outgoing (Reverse) Connection









Task 3 – Process Investigation\

```
┌──(kali㉿kali)-[~]
└─$ ps -o pid,ppid,user,cmd -p 20
    PID    PPID USER     CMD
     20       2 root     [cpuhp/0]
```

```
┌──(kali㉿kali)-[~]
└─$ pgrep -a sshd

┌──(kali㉿kali)-[~]
└─$ ps -o pid,ppid,user -p 3412
    PID    PPID USER

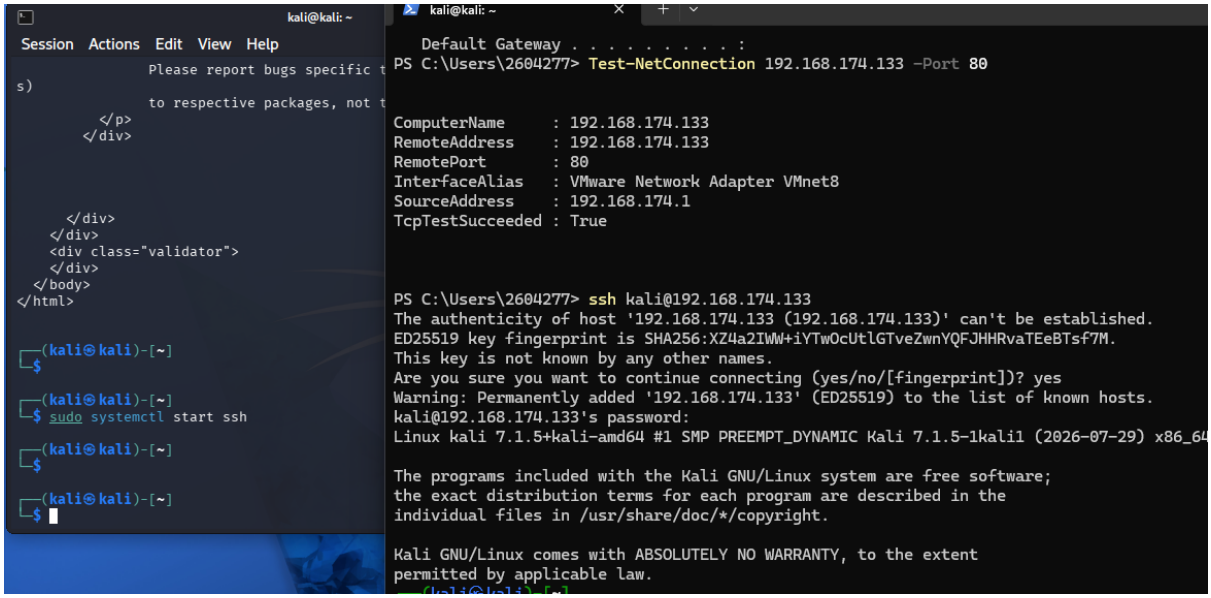
┌──(kali㉿kali)-[~]
└─$ sudo readlink /proc/3412/exe
```

```
┌──(kali㉿kali)-[~]
└─$ sudo cat /proc/20/cmdline | tr '/D' ' '; echo


┌──(kali㉿kali)-[~]
└─$ sudo ss -tlnp | grep 'pid=20'
```

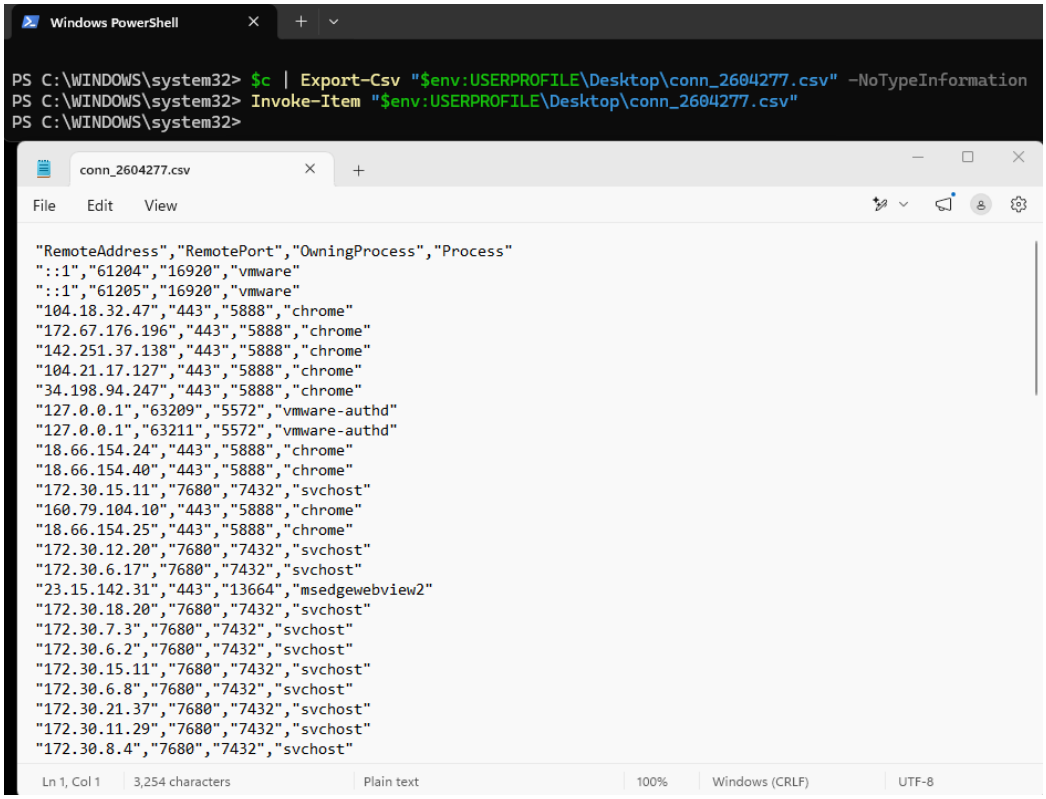
Task 4 – Windows Connection Triage

I started ssh in kali and access it from windows:




```
PS C:\WINDOWS\system32> $c = Get-NetTCPConnection -State Established |
>> Select-Object RemoteAddress, RemotePort, OwningProcess,
>> @{Name='Process';Expression={
>>     (Get-Process -Id $_.OwningProcess -ErrorAction SilentlyContinue).ProcessName
>> }}
PS C:\WINDOWS\system32>
PS C:\WINDOWS\system32> $c | Format-Table -AutoSize

RemoteAddress   RemotePort OwningProcess Process
-------------   ---------- ------------- -------
::1                  61204         16920 vmware
::1                  61205         16920 vmware
104.18.32.47           443          5888 chrome
172.67.176.196         443          5888 chrome
142.251.37.138         443          5888 chrome
104.21.17.127          443          5888 chrome
34.198.94.247          443          5888 chrome
127.0.0.1            63209          5572 vmware-authd
127.0.0.1            63211          5572 vmware-authd
18.66.154.24           443          5888 chrome
18.66.154.40           443          5888 chrome
172.30.15.11          7680          7432 svchost
160.79.104.10          443          5888 chrome
18.66.154.25           443          5888 chrome
172.30.12.20          7680          7432 svchost
172.30.6.17           7680          7432 svchost
23.15.142.31           443         13664 msedgewebview2
```



Saving and output

Task 5 – Sysmon Detection

Understanding how to complete this task. The process involves ensuring Sysmon is installed and running on Windows, starting Apache on Kali Linux, generating an HTTP request from Windows using PowerShell's Invoke-WebRequest, and then examining Sysmon Event ID 3 (Network Connection) and Event ID 1 (Process Creation) using Get-WinEvent. The events are used to identify the executed command, parent process, destination IP address, and destination port.

Issue faced: I successfully started Apache on Kali Linux and generated an HTTP request from Windows. The request was received successfully, as confirmed by Kali's Apache access log (HTTP 200 response). However, the university Windows environment does not have Sysmon installed, and I do not have administrator privileges to install or configure it. Consequently, the Microsoft-Windows-Sysmon/Operational event log is unavailable, preventing me from retrieving the required events.

All previous lab tasks have been completed. Task 5 remains partially completed due to this environment limitation.

Required to complete: Access to a Windows environment with administrator privileges, Sysmon installed and configured with Event ID 3 enabled, and permission to read the Sysmon Operational event log.